# Lecture 4

Saturday, February 14, 2026 9:13 AM

Why study probability ?

ML < data incomplete / noisy data } uncertainty

Probability theory provides a principled way to deal with uncertainty.
Quantify our confidence in our predictions.

Fundamentals

1. Random experiment
Example : Flipping a coin
Rolling a dice

2. Outcome of an experiment
The result of a single run of an experiment
Example : Flipping a coin — heads / tails
Rolling a dice — 1, 2, 3, 4, 5, 6

3. Sample space ($\Omega$) : The set of all possible outcomes of a random experiment
Example: Coin toss = {heads, tails}
Dice = {1, 2, 3, 4, 5, 6}

4. Event : Subsets of $\Omega$ (sample space)
Example : Coin : getting a head = {heads}
Dice : a) even number = {2, 4, 6} = A
b) prime number = {2, 3, 5} = B

A & B

$\underbrace{A \cup B}_{\text{either A or B or both occur}}$ , $\underbrace{A \cap B}$ → A & B both occur

$A \cup B = \{2, 3, 4, 5, 6\}$
$A \cap B = \{2\}$

Computing probabilityies when all outcomes are equally likely

Coin toss : fair coin

$$\mathbb{P}(\text{heads}) = \frac{\text{getting heads}}{\underset{\hookrightarrow \text{ \# of elements in } \Omega}{|\Omega|}} = \frac{|\{\text{heads}\}|}{|\Omega|} = \frac{1}{2}$$

Rolling a dice
$A = \{2, 4, 6\}$ $\quad \mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{1}{2}$
$\Omega = \{1, 2, 3, 4, 5, 6\}$

$B = \{2, 3, 5\}$ $\quad \mathbb{P}(B) = \frac{|B|}{|\Omega|} = \frac{1}{2}$

$A \cap B = \{2\}$ $\quad \mathbb{P}(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{1}{6}$

Bayes' theorem

Updating belief based on additional information.

Random experiment, $\Omega$, events A & B, $\mathbb{P}(B) > 0$.

$$\underbrace{\mathbb{P}(A \mid B)}_{\text{Event A, given event B has occured}} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \text{— } *$$

A : {2, 4, 6} $\quad \mathbb{P}(A) = 1/2$ — ①
B : {2, 3, 5} $\quad \mathbb{P}(B) = 1/2$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/6}{1/2} = 1/3 \quad \text{— ②}$$

Independence
(of events)

A, B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$. — (*)

Plug this back into Bayes' theorem

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A)\cancel{\mathbb{P}(B)}}{\cancel{\mathbb{P}(B)}} \quad (\text{by } *)$$

$\underbrace{\mathbb{P}(A \mid B) = \mathbb{P}(A)}$ — what does this mean?

A : {2, 4, 6} $\quad$ Is A independent of B ?
B : {2, 3, 5}
(Recall $\mathbb{P}(A \mid B) \neq \mathbb{P}(A)$)

If we toss a fair coin twice

C : get a heads in the first toss
D : get a tails in the second toss

Is C independent of D ?

{HH, HT, TH, TT} $\Omega$

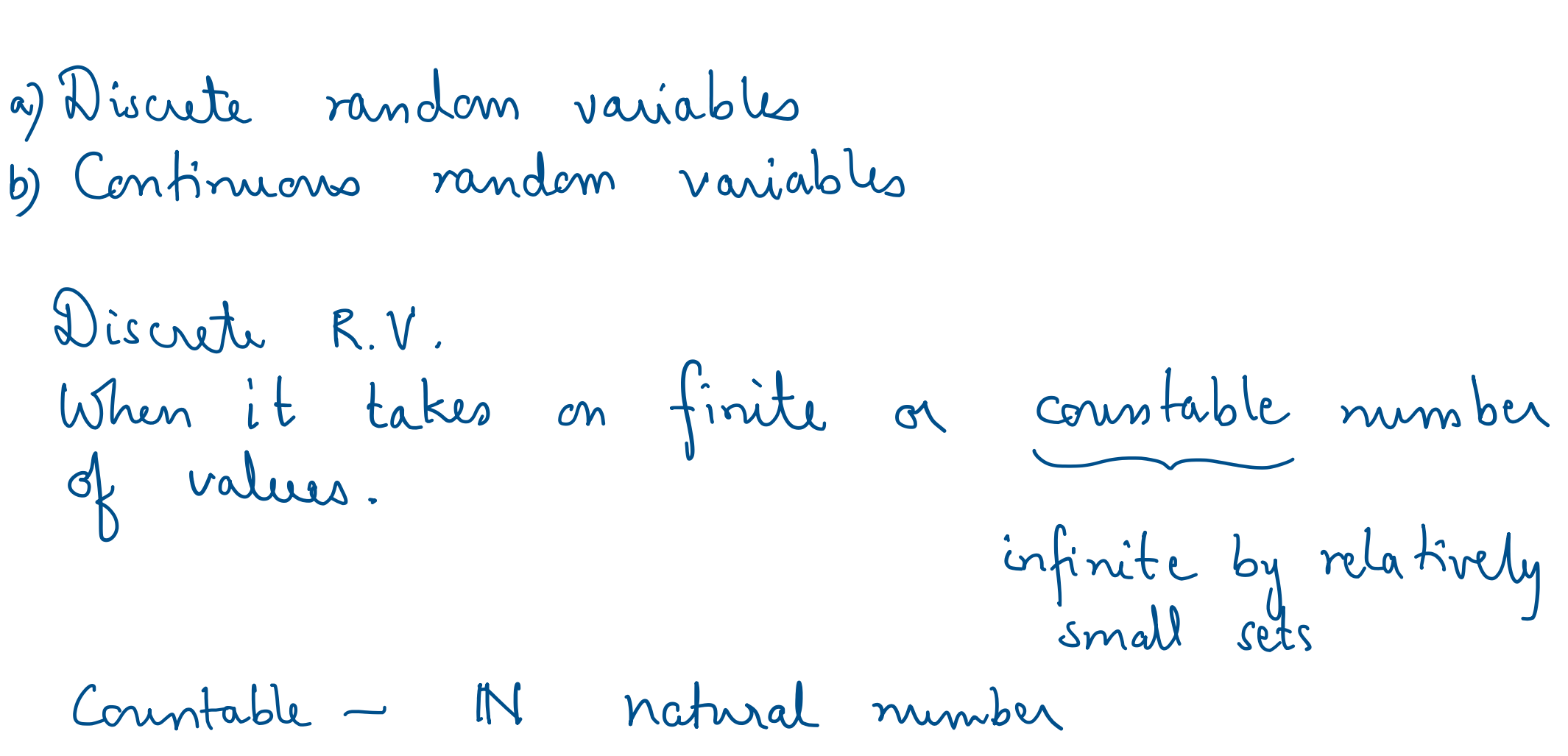
$C = \{HH, HT\}$ $\quad \mathbb{P}(C) = 1/2$
$D = \{HT, TT\}$ $\quad \mathbb{P}(D) = 1/2$
$C \cap D = \{HT\}$ $\quad \mathbb{P}(C \cap D) = \frac{|C \cap D|}{|\Omega|} = 1/4$

$\mathbb{P}(C \cap D) = \mathbb{P}(C)\,\mathbb{P}(D)$

Random Variables
Deterministic function
Misnomer

Function from $\Omega$ to $\mathbb{R}$
$\Omega$ ↓ sample space
$\mathbb{R}$ ↪ real line

toss a coin 3 times



Why do we need random variables ?
- We can do mathematical operations on outcomes
- Infer the statistics of the outcome

a) Discrete random variables
b) Continuous random variables

Discrete R.V.
When it takes on finite or $\underbrace{\text{countable}}_{\text{infinite by relatively small sets}}$ number of values.

Countable — $\mathbb{N}$ natural number
$\mathbb{Z}$ integers
$\mathbb{Q}$ rational number

PMF
Probability Mass function

$$\underset{\substack{\uparrow \\ \text{random variable}}}{p_X}(x) = \mathbb{P}(X = x) \quad \leftarrow \text{numbers}$$

Properties :
① $p_X(x) \geq 0 \quad \forall x$
② $\sum_x p_X(x) = 1$

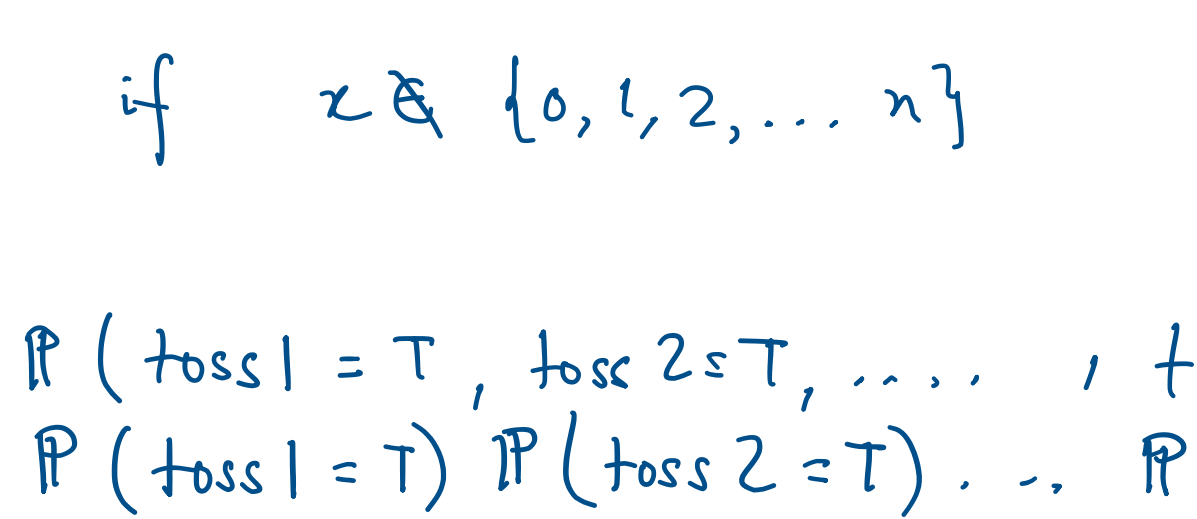
Corollory $\quad p_X(x) > 1$ ? $\quad$ Not possible

Examples :

① Bernoulli

$$X = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

Interpretation : Toss of toss with $\mathbb{P}(\text{heads}) = p$



② Binomial random variable

$X \sim \text{Bin}(n, p)$
n ↓ natural number
p ↪ $\in [0, 1]$

Interpretation : # of heads you get if you toss a coin with $\mathbb{P}(\text{heads}) = p$, n times

X takes on values in {0, 1, 2, ... n}

$$p_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x \in \{0, 1, 2, \dots, n\} \\ 0 & \text{if } x \notin \{0, 1, 2, \dots n\} \end{cases}$$

$p_X(0) = (1-p)^n$
(calculation : $\mathbb{P}(\text{toss } 1 = T, \text{toss } 2 = T, \dots, \text{toss } n = T)$
$= \mathbb{P}(\text{toss } 1 = T)\,\mathbb{P}(\text{toss } 2 = T) \dots \mathbb{P}(\text{toss } n = T)$
$= (1-p)^n$)

$p_X(1) = \binom{n}{1} p (1-p)^{n-1}$

(Calculation for N = 3)
$p_X(1) = \mathbb{P}(HTT, THT, TTH)$
$= \mathbb{P}(HTT) + \mathbb{P}(THT) + \mathbb{P}(TTH)$
$= p(1-p)^2 + (1-p)p(1-p) + (1-p)^2 p$
$= 3p(1-p)^2$

$p_X(x)$ for $x \in \{0, 1, 2, \dots n\}$

$= \binom{n}{x} p^x (1-p)^{n-x}$ ←

③ Geometric random variable
X = # of coin tosses needed to get the first heads

X can take the values {1, 2, .... } $\mathbb{N}$

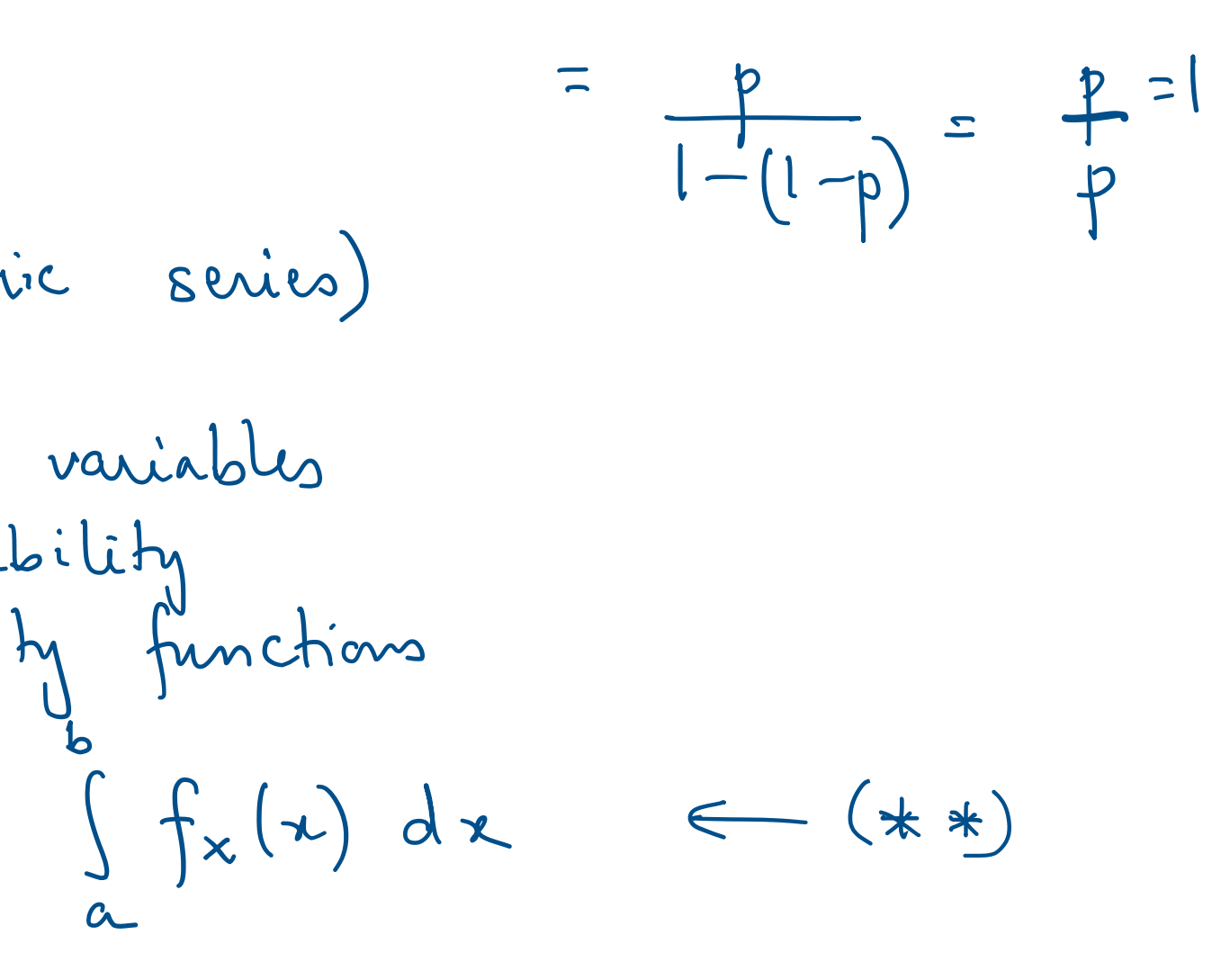
$p_X(x) = $ ?

$x = 1 \quad p_X(1) = p$
$x = 2 \quad p_X(2) = (1-p)\,p$
⋮
$x = k \quad p_X(k) = (1-p)^{k-1} p$ ←

$X = k \quad \underbrace{TTT \dots T}_{k-1} H$

Sanity check : $\sum_{x=1}^{\infty} p_X(x) = $ ?

Sample space



$p_X(x) = (1-p)^{x-1} p \qquad x \in \mathbb{N}$

$$\sum_{x=1}^{\infty} p_X(x) = \sum_{x=1}^{\infty} (1-p)^{x-1} p = p \sum_{x=0}^{\infty} (1-p)^x$$

$$= \frac{p}{1-(1-p)} = \frac{p}{p} = 1$$

(Sum a Geometric series)

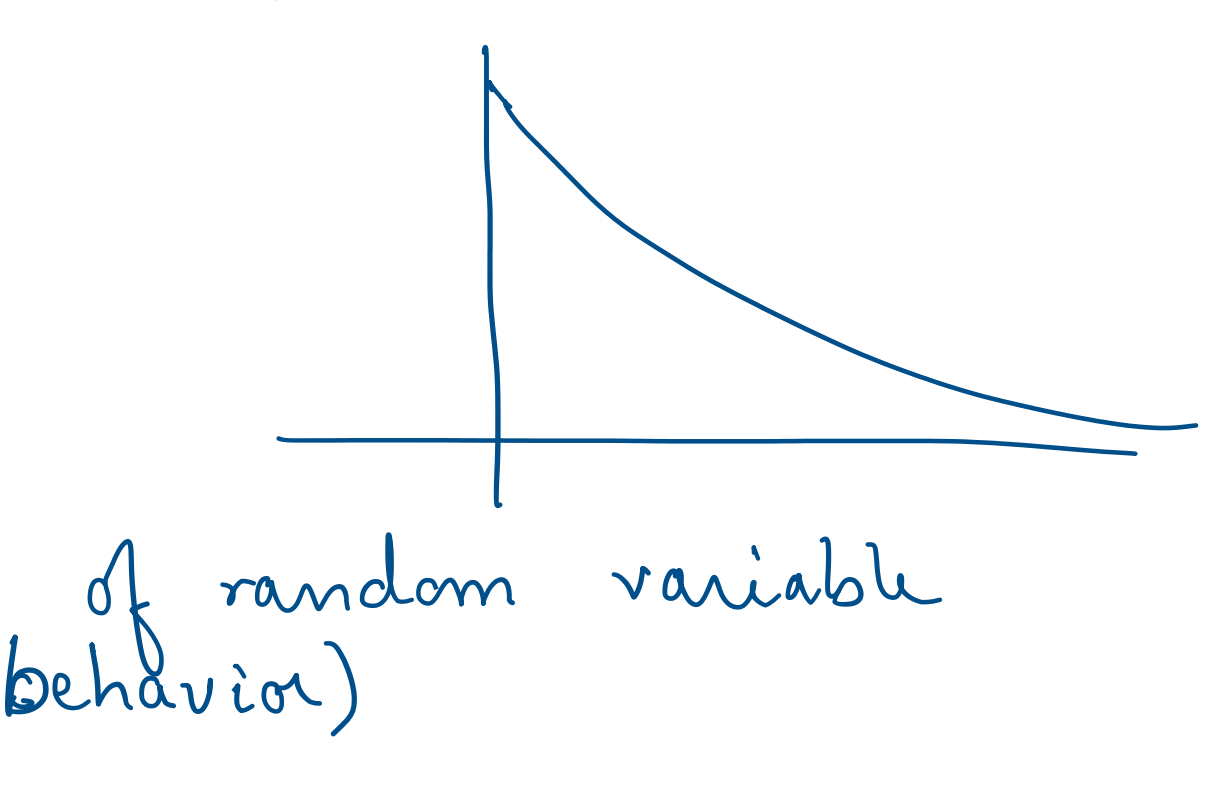
Continuous random variables

$f_X(x) \leftarrow$ probability density functions

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\,dx \quad \leftarrow (**)$$

Gaussian random variable

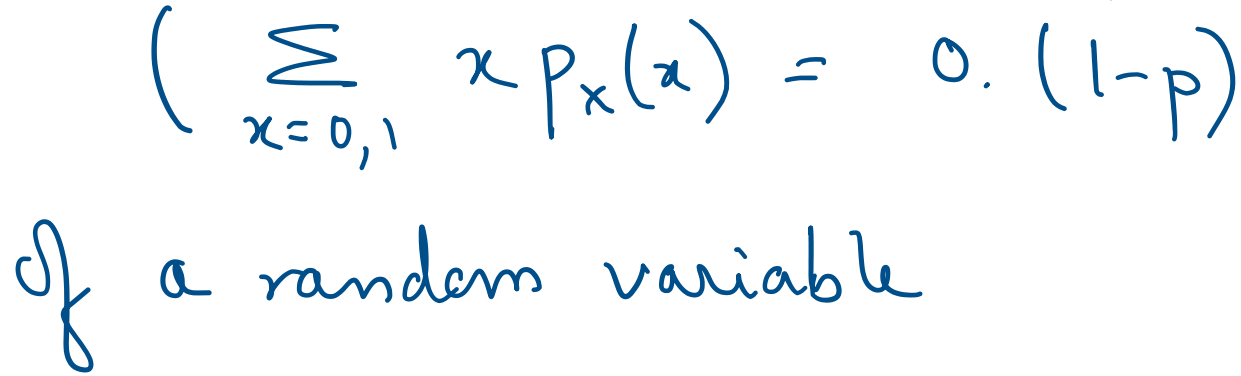
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \forall x$$



$\mathbb{P}(X = 5) = \int_5^5 f_X(x)\,dx = 0$

Exponential

$f_X(x) = \lambda e^{-\lambda x} \qquad x \geq 0$



- Expection of random variable
(average behavior)

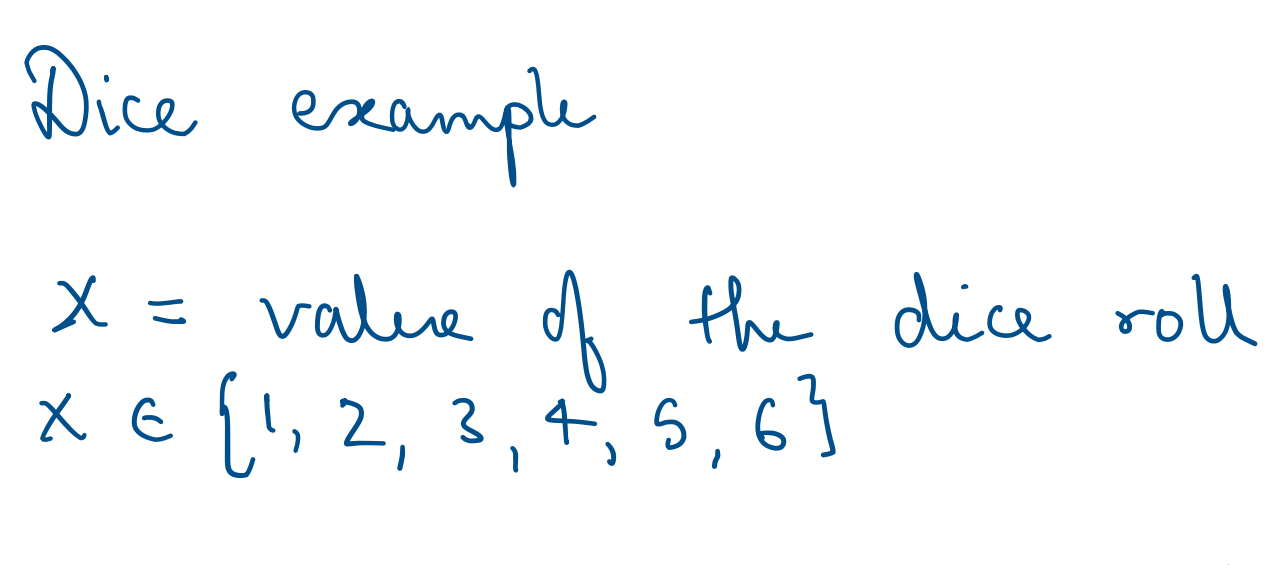
$\mathbb{E}[X] = \sum_x x\,p_X(x)$ $\quad$ for discrete

$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$ $\quad$ for continuous

$X \sim \text{Ber}(p)$ , recall $X = \begin{cases} 0 & \text{w.p. } 1-p \\ 1 & \text{w.p. } p \end{cases}$

$\mathbb{E}[X] = p \quad \left(\sum_{x=0,1} x\,p_X(x) = 0.(1-p) + 1.p = p\right)$

- Variance of a random variable



$\mathbb{E}[X] = 0$ $\qquad$ $\mathbb{E}[Y] = 0$

$$\text{var}(X) = \mathbb{E}\left[(X - \underset{\substack{\downarrow \\ \text{expected value}}}{\mathbb{E}[X]})^2\right]$$

Conditional Expectation (more later)

Dice example

X = value of the dice roll
$X \in \{1, 2, 3, 4, 5, 6\}$

$\mathbb{E}[X] = \frac{1}{6}[1+2+3+4+5+6] = 3.5$

A = we get an even number

$\mathbb{E}[X \mid A] = \frac{1}{3}[2+4+6] = 4$
↑ random variable ↑ event

notation
↓
$p_{X|A}(2) = 1/3$
$p_{X|A}(4) = 1/3$
$p_{X|A}(6) = 1/3$

$\mathbb{E}[X \mid A^c] = \frac{1}{3}[1+3+5] = 3$

$p_{X|A^c}(1) = 1/3$
$p_{X|A^c}(3) = 1/3$
$p_{X|A^c}(5) = 1/3$

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